

**CONFIDENTIAL**



**UNIVERSITI TEKNOLOGI MALAYSIA**  
**FINAL EXAMINATION**  
**SEMESTER II 2022/2023**

**SUBJECT CODE : SCSI/SECI 1143/2143**  
**SUBJECT NAME : PROBABILITY & STATISTICAL DATA ANALYSIS**  
**YEAR/COURSE :**  
**TIME :**  
**DATE :**  
**VENUE :**

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**INSTRUCTIONS TO THE STUDENTS:**

Answer all questions in the answer booklet.

Do all calculations in 3 decimal points.

|                   |  |
|-------------------|--|
| <b>NAME</b>       |  |
| <b>MATRIC NO.</b> |  |
| <b>SECTION</b>    |  |
| <b>LECTURER</b>   |  |

(This question paper consist of 15 pages including this pages)

## QUESTION 1

[10 MARKS]

- a) A recent study investigated the usage of ChatGPT towards the students. ChatGPT is a tool that provide language learning, translation and many more. Researcher observed that a representative sample of students from Faculty of Computing found that 158 out of 367 students are active using the ChatGPT.
- i) Compute a point estimate of the true proportion of all students who active using the ChatGPT.  
(1 mark)
  - ii) Construct and interpret a 95% confidence interval for the true proportion of all students who actively using ChatGPT services.  
(2 marks)
- b) In a study of coffee lover among staff in UTM, 30 randomly selected staffs were asked how much time they drink coffee in a day. The time is assumed normally distributed and the sample standard deviation was 6 hours. At  $\alpha = 0.01$ , is there enough evidence to reject the claim that the population variance is at 35 hours.
- i) State the null and alternative hypothesis.  
(1 mark)
  - ii) Calculate the test statistic value.  
(2 marks)
  - iii) Find the critical value.  
(1 mark)
  - iv) Draw and shade the critical region (rejection area) and what can you conclude from the test.  
(2 marks)

**QUESTION 2****[20 MARKS]**

- a) A research study was conducted to examine the differences between older and younger adults on perceived life satisfaction. A pilot study was conducted to examine this hypothesis. 13 older adults (over the age of 70) and 16 younger adults (between 20 and 30) were give a life satisfaction test. The summary of the score results for the test is given in Table 1.

**Table 1:** Data on perceived life satisfaction

| Older adults | Younger adults |
|--------------|----------------|
| $s_1=1.21$   | $s_2=0.63$     |

Test at the  $\alpha = 0.05$  level of significance whether there is a different in the variability between the two groups?

(10 marks)

- b) The iron contents of fruits before and after applying farm yard manure were observed as in Table 2.

**Table 2:** The iron contents of fruits before and after applying farm yard manure

| Fruit No        | 1    | 2    | 3    | 4    | 5    | 6    | 7   | 8    | 9    |
|-----------------|------|------|------|------|------|------|-----|------|------|
| Before Applying | 11.6 | 18.7 | 15.9 | 20.6 | 10.1 | 17.4 | 7.2 | 12.2 | 11.7 |
| After Applying  | 10.0 | 21.6 | 13.9 | 22.8 | 11.5 | 16.2 | 8.1 | 10.8 | 9.6  |

Is there any significant difference between the mean iron contents in the fruits before and after the farm yarn manure at  $\alpha = 0.05$  level of significance?

(10 marks)

**QUESTION 3****[20 MARKS]**

- a) A staff member of an emergency medical service wishes to determine whether the number of accidents is equally distributed during a week. A week was selected at random, and the following data (Table 3) were obtained. Is there evidence to reject the hypothesis that the number of accident is equally distributed throughout the week, at 95% confidence level.

**Table 3:** Distribution number of accident during a week

| Day             | Mon. | Tues. | Wed. | Thurs. | Fri. | Sat. | Sun. |
|-----------------|------|-------|------|--------|------|------|------|
| No. of accident | 28   | 21    | 15   | 14     | 32   | 23   | 19   |

(8 marks)

- b) A study is being conducted to determine whether there is relationship between jogging and blood pressure. A random sample of 210 subjects is selected, and the raw data is shown in Table 4.

**Table 4:** Data from 210 respondents

| Jogging Status | Blood pressure level | Count |
|----------------|----------------------|-------|
| Joggers        | Low                  | 34    |
| Non joggers    | Low                  | 15    |
| Joggers        | Moderate             | 57    |
| Non joggers    | Moderate             | 63    |
| Joggers        | High                 | 21    |
| Non joggers    | High                 | 20    |

Based on the data,

- i) Construct the two ways contingency table based on data in Table 4.

(2 marks)

- ii) At  $\alpha = 0.05$  level of significance, test the claim that jogging and blood pressure are not related.

(10 marks)

**QUESTION 4****[15 MARKS]**

A study of the amount of rainfall and the quantity of air pollution removed produced the following data (Table 5). Assume that  $x$  and  $y$  are random variables with a bivariate normal distribution.

**Table 5:** Rainfall and air pollution data

| Daily rainfall, $x$ (0.01 cm) | Particulate removed, $y$ ( $\frac{\mu g}{m^3}$ ) |
|-------------------------------|--------------------------------------------------|
| 4.3                           | 126                                              |
| 4.5                           | 121                                              |
| 5.9                           | 116                                              |
| 5.6                           | 118                                              |
| 6.1                           | 114                                              |
| 5.2                           | 118                                              |
| 3.8                           | 132                                              |
| 2.1                           | 141                                              |
| 7.5                           | 108                                              |

a) Draw (sketch) the scatter plot.

(2 marks)

b) Given that  $\sum x^2 = 244.26$ ,  $\sum y^2 = 133786$ , calculate  $r$ .

(10 marks)

c) Does the value of  $r$  value found in 4(b) consistent with the pattern of the scatter plot in 4(a)?

(1 mark)

d) Is the  $r$  value found in 4(b) suggest a strong linear relationship? If yes, do the changes in  $x$  and  $y$  will go in the same direction?

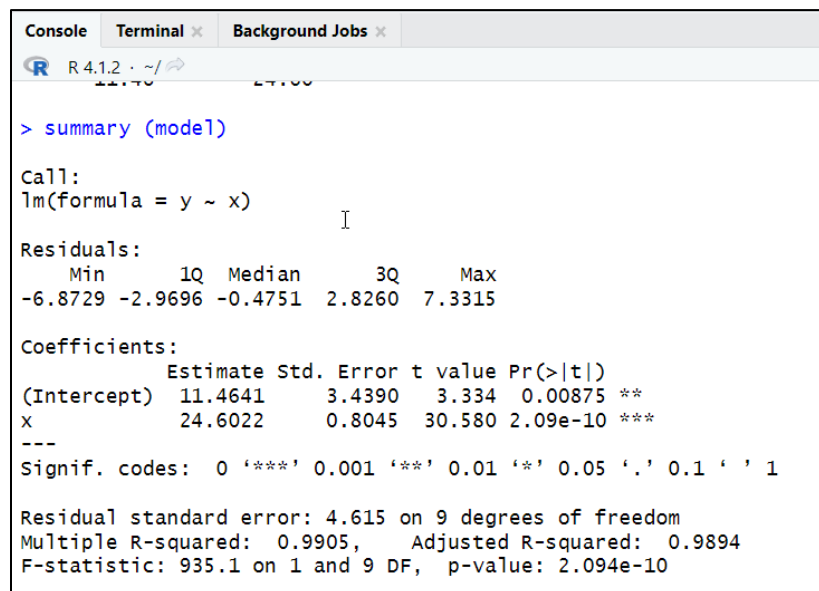
(2 marks)

**QUESTION 5****[15 MARKS]**

Accu-Copiers, Inc., sells and services the Accu-500 copying machine. As part of its standard service contract, the company agrees to perform routine service on this copier. To obtain information about the time it takes to perform routine service, Accu-Copiers has collected data for 11 service calls. The data are given in Table 6. Whilst, the R output from fitting a least squares regression line to the data is shown in Figure 1.

**Table 6:** The service time data

| Service Call | Number of copiers serviced, $x$ | Number of minutes required, $y$ |
|--------------|---------------------------------|---------------------------------|
| 1            | 4                               | 109                             |
| 2            | 2                               | 58                              |
| 3            | 5                               | 138                             |
| 4            | 7                               | 189                             |
| 5            | 1                               | 37                              |
| 6            | 3                               | 82                              |
| 7            | 4                               | 103                             |
| 8            | 5                               | 134                             |
| 9            | 2                               | 68                              |
| 10           | 4                               | 112                             |
| 11           | 6                               | 154                             |



```
Console Terminal x Background Jobs x
R 4.1.2 ~ /
> summary(model)

Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-6.8729 -2.9696 -0.4751  2.8260  7.3315

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.4641     3.4390   3.334  0.00875 **
x             24.6022     0.8045  30.580 2.09e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.615 on 9 degrees of freedom
Multiple R-squared:  0.9905,    Adjusted R-squared:  0.9894
F-statistic: 935.1 on 1 and 9 DF,  p-value: 2.094e-10
```

**Figure 1:** R Output

- a) Find the least squares point estimates  $b_0$  and  $b_1$  on the R output (Figure 1) and write their value.  
(2 marks)
- b) Interpret  $b_0$  and  $b_1$ . Does the interpretation of  $b_0$  make practical sense?  
(3 marks)
- c) Write the equation of the least squares line.  
(2 marks)
- d) Use the least square line to compute a point estimate of the mean time to service four copiers.  
(2 marks)
- e) How many percent of the variation in the number of minutes required to service is explained by the variation in the number of copiers serviced?  
(1 mark)
- f) Conduct the  $t$ -test to check whether there is a linear relationship between  $x$  and  $y$ . Use  $\alpha = 0.05$   
(5 marks)

**QUESTION 6****[20 MARKS]**

A scientist wanted to see if mice is sensitive to colour. He set up an experiment using mice and different door colours to see if there is any colour that the mice would be attracted to. Three sets of five mice were randomly selected to be placed in a standard maze but with different colour doors: red, green, and black. The objective of the experiment is to identify the time for each mouse to run in the maze and reach the door. He recorded the response time (in minutes), to which the time required for each mouse to complete the maze (Table 7).

**Table 7:** Response time recorded for each mouse to complete the maze.

| Door Colour | Response time (minutes) |    |    |    |    |
|-------------|-------------------------|----|----|----|----|
|             | M1                      | M2 | M3 | M4 | M5 |
| Red         | 9                       | 11 | 10 | 9  | 15 |
| Green       | 20                      | 21 | 23 | 17 | 30 |
| Black       | 6                       | 5  | 8  | 14 | 7  |

- a) Find the mean,  $\bar{x}$ , and standard deviation,  $s$ , for each door colour.

(6 marks)

- b) Calculate the test statistics,  $F$ .

(10 marks)

- c) At the 0.01 level of significance, test if the data provide sufficient evidence to conclude that mice are sensitive to different colour.

(4 marks)

### List of Formula:

(a) Point Estimate and CI

$$\hat{p} \pm (z \text{ critical value}) \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

(b) Hypothesis Testing for one sample

$$\text{Test Statistic, } z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$\text{Test Statistic, } t_0 = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$$\text{Test Statistic, } \chi^2 = \frac{(n - 1)s^2}{\sigma^2}$$

(c) Hypothesis Testing for two samples

$$\text{Test Statistic, } z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\text{Test Statistic, } t_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

$$\text{Test Statistic, } F_0 = \frac{s_1^2}{s_2^2}$$

(d) Chi-Square Test

$$\text{Test Statistic, } \chi^2 = \sum_i^n \frac{(o_i - e_i)^2}{e_i}$$

$$\text{Degree of Freedom for two - way contingency table} = (r - 1)(c - 1)$$

(e) Correlation

$$\text{Pearson Correlation Coefficient, } r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$\text{Spearman Correlation Coefficient, } r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \text{ where } d_i = y_i - x_i$$

$$\text{Test Statistic, } t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}}$$

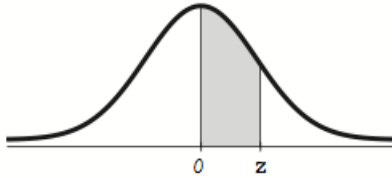
(f) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

(g) Analysis of variance (ANOVA)

$$\text{Test Statistic, } F = \frac{ns^{\frac{2}{x}}}{s_p^2}$$



**TABLE A-2** Standard Normal (z) Distribution

| z               | .00   | .01   | .02   | .03   | .04   | .05    | .06   | .07   | .08    | .09   |
|-----------------|-------|-------|-------|-------|-------|--------|-------|-------|--------|-------|
| 0.0             | .0000 | .0040 | .0080 | .0120 | .0160 | .0199  | .0239 | .0279 | .0319  | .0359 |
| 0.1             | .0398 | .0438 | .0478 | .0517 | .0557 | .0596  | .0636 | .0675 | .0714  | .0753 |
| 0.2             | .0793 | .0832 | .0871 | .0910 | .0948 | .0987  | .1026 | .1064 | .1103  | .1141 |
| 0.3             | .1179 | .1217 | .1255 | .1293 | .1331 | .1368  | .1406 | .1443 | .1480  | .1517 |
| 0.4             | .1554 | .1591 | .1628 | .1664 | .1700 | .1736  | .1772 | .1808 | .1844  | .1879 |
| 0.5             | .1915 | .1950 | .1985 | .2019 | .2054 | .2088  | .2123 | .2157 | .2190  | .2224 |
| 0.6             | .2257 | .2291 | .2324 | .2357 | .2389 | .2422  | .2454 | .2486 | .2517  | .2549 |
| 0.7             | .2580 | .2611 | .2642 | .2673 | .2704 | .2734  | .2764 | .2794 | .2823  | .2852 |
| 0.8             | .2881 | .2910 | .2939 | .2967 | .2995 | .3023  | .3051 | .3078 | .3106  | .3133 |
| 0.9             | .3159 | .3186 | .3212 | .3238 | .3264 | .3289  | .3315 | .3340 | .3365  | .3389 |
| 1.0             | .3413 | .3438 | .3461 | .3485 | .3508 | .3531  | .3554 | .3577 | .3599  | .3621 |
| 1.1             | .3643 | .3665 | .3686 | .3708 | .3729 | .3749  | .3770 | .3790 | .3810  | .3830 |
| 1.2             | .3849 | .3869 | .3888 | .3907 | .3925 | .3944  | .3962 | .3980 | .3997  | .4015 |
| 1.3             | .4032 | .4049 | .4066 | .4082 | .4099 | .4115  | .4131 | .4147 | .4162  | .4177 |
| 1.4             | .4192 | .4207 | .4222 | .4236 | .4251 | .4265  | .4279 | .4292 | .4306  | .4319 |
| 1.5             | .4332 | .4345 | .4357 | .4370 | .4382 | .4394  | .4406 | .4418 | .4429  | .4441 |
| 1.6             | .4452 | .4463 | .4474 | .4484 | .4495 | *.4505 | .4515 | .4525 | .4535  | .4545 |
| 1.7             | .4554 | .4564 | .4573 | .4582 | .4591 | ↑.4599 | .4608 | .4616 | .4625  | .4633 |
| 1.8             | .4641 | .4649 | .4656 | .4664 | .4671 | ↑.4678 | .4686 | .4693 | .4699  | .4706 |
| 1.9             | .4713 | .4719 | .4726 | .4732 | .4738 | ↑.4744 | .4750 | .4756 | .4761  | .4767 |
| 2.0             | .4772 | .4778 | .4783 | .4788 | .4793 | .4798  | .4803 | .4808 | .4812  | .4817 |
| 2.1             | .4821 | .4826 | .4830 | .4834 | .4838 | .4842  | .4846 | .4850 | .4854  | .4857 |
| 2.2             | .4861 | .4864 | .4868 | .4871 | .4875 | .4878  | .4881 | .4884 | .4887  | .4890 |
| 2.3             | .4893 | .4896 | .4898 | .4901 | .4904 | .4906  | .4909 | .4911 | .4913  | .4916 |
| 2.4             | .4918 | .4920 | .4922 | .4925 | .4927 | .4929  | .4931 | .4932 | .4934  | .4936 |
| 2.5             | .4938 | .4940 | .4941 | .4943 | .4945 | .4946  | .4948 | .4949 | *.4951 | .4952 |
| 2.6             | .4953 | .4955 | .4956 | .4957 | .4959 | .4960  | .4961 | .4962 | ↑.4963 | .4964 |
| 2.7             | .4965 | .4966 | .4967 | .4968 | .4969 | .4970  | .4971 | .4972 | ↑.4973 | .4974 |
| 2.8             | .4974 | .4975 | .4976 | .4977 | .4977 | .4978  | .4979 | .4979 | .4980  | .4981 |
| 2.9             | .4981 | .4982 | .4982 | .4983 | .4984 | .4984  | .4985 | .4985 | .4986  | .4986 |
| 3.0             | .4987 | .4987 | .4987 | .4988 | .4988 | .4989  | .4989 | .4989 | .4990  | .4990 |
| 3.10 and higher | .4999 |       |       |       |       |        |       |       |        |       |

NOTE: For values of z above 3.09, use 0.4999 for the area.

\*Use these common values that result from interpolation:

z score      Area

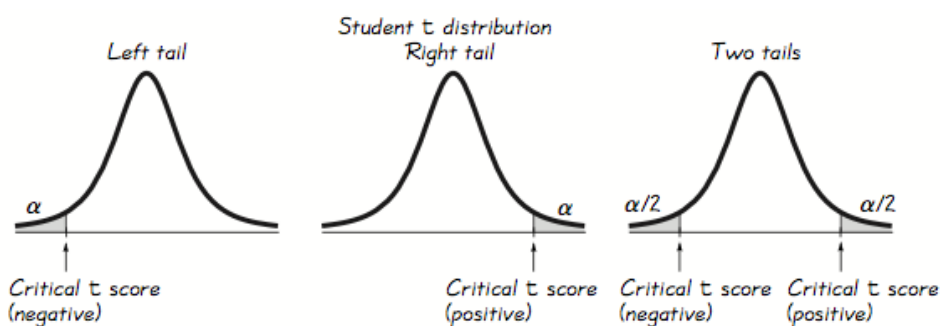
1.645      0.4500      ←

2.575      0.4950      ←

From Frederick C. Mosteller and Robert E. K. Rourke, *Sturdy Statistics*, 1973, Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of Frederick Mosteller.

| STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score. |        |        |        |        |        |        |        |        |        |        |  |  |  |  |  |
|---------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--|--|--|--|--|
| Z                                                                                     | .00    | .01    | .02    | .03    | .04    | .05    | .06    | .07    | .08    | .09    |  |  |  |  |  |
| -3.9                                                                                  | .0005  | .0005  | .0004  | .0004  | .0004  | .0004  | .0004  | .0004  | .0003  | .0003  |  |  |  |  |  |
| -3.8                                                                                  | .0007  | .0007  | .0007  | .0006  | .0006  | .0006  | .0006  | .0005  | .0005  | .0005  |  |  |  |  |  |
| -3.7                                                                                  | .0011  | .0011  | .0010  | .0010  | .0009  | .0009  | .0008  | .0008  | .0008  | .0008  |  |  |  |  |  |
| -3.6                                                                                  | .0016  | .0015  | .0015  | .0014  | .0014  | .0013  | .0013  | .0012  | .0012  | .0011  |  |  |  |  |  |
| -3.5                                                                                  | .0023  | .0022  | .0022  | .0021  | .0020  | .0019  | .0019  | .0018  | .0017  | .0017  |  |  |  |  |  |
| -3.4                                                                                  | .0034  | .0032  | .0031  | .0030  | .0029  | .0028  | .0027  | .0026  | .0025  | .0024  |  |  |  |  |  |
| -3.3                                                                                  | .0048  | .0047  | .0045  | .0043  | .0042  | .0040  | .0039  | .0038  | .0036  | .0035  |  |  |  |  |  |
| -3.2                                                                                  | .0069  | .0066  | .0064  | .0062  | .0060  | .0058  | .0056  | .0054  | .0052  | .0050  |  |  |  |  |  |
| -3.1                                                                                  | .0099  | .0094  | .0090  | .0087  | .0084  | .0082  | .0079  | .0076  | .0074  | .0071  |  |  |  |  |  |
| -3.0                                                                                  | .0135  | .0131  | .0126  | .0122  | .0118  | .0114  | .0111  | .0107  | .0104  | .0101  |  |  |  |  |  |
| -2.9                                                                                  | .0187  | .0181  | .0175  | .0169  | .0164  | .0159  | .0154  | .0149  | .0144  | .0139  |  |  |  |  |  |
| -2.8                                                                                  | .0256  | .0248  | .0240  | .0233  | .0226  | .0219  | .0212  | .0205  | .0199  | .0193  |  |  |  |  |  |
| -2.7                                                                                  | .0347  | .0336  | .0326  | .0317  | .0307  | .0298  | .0289  | .0280  | .0272  | .0264  |  |  |  |  |  |
| -2.6                                                                                  | .0466  | .0453  | .0440  | .0427  | .0415  | .0402  | .0391  | .0379  | .0368  | .0357  |  |  |  |  |  |
| -2.5                                                                                  | .0621  | .0604  | .0587  | .0570  | .0554  | .0539  | .0523  | .0508  | .0494  | .0480  |  |  |  |  |  |
| -2.4                                                                                  | .0820  | .0798  | .0776  | .0755  | .0734  | .0714  | .0695  | .0676  | .0657  | .0639  |  |  |  |  |  |
| -2.3                                                                                  | .1072  | .1044  | .1017  | .0990  | .0964  | .0939  | .0914  | .0889  | .0866  | .0842  |  |  |  |  |  |
| -2.2                                                                                  | .1390  | .1355  | .1321  | .1287  | .1255  | .1222  | .1191  | .1160  | .1130  | .1101  |  |  |  |  |  |
| -2.1                                                                                  | .1786  | .1743  | .1700  | .1659  | .1618  | .1578  | .1539  | .1500  | .1463  | .1426  |  |  |  |  |  |
| -2.0                                                                                  | .2275  | .2222  | .2169  | .2118  | .2068  | .2018  | .2018  | .2018  | .2018  | .2018  |  |  |  |  |  |
| -1.9                                                                                  | .2872  | .2807  | .2743  | .2680  | .2619  | .2559  | .2500  | .2442  | .2385  | .2330  |  |  |  |  |  |
| -1.8                                                                                  | .3593  | .3515  | .3438  | .3362  | .3288  | .3216  | .3144  | .3074  | .3005  | .2938  |  |  |  |  |  |
| -1.7                                                                                  | .4457  | .4363  | .4272  | .4182  | .4093  | .4006  | .3920  | .3836  | .3754  | .3673  |  |  |  |  |  |
| -1.6                                                                                  | .5480  | .5370  | .5262  | .5155  | .5050  | .4947  | .4846  | .4746  | .4648  | .4551  |  |  |  |  |  |
| -1.5                                                                                  | .6681  | .6552  | .6426  | .6301  | .6178  | .6057  | .5938  | .5821  | .5705  | .5592  |  |  |  |  |  |
| -1.4                                                                                  | .8076  | .7927  | .7780  | .7636  | .7493  | .7353  | .7215  | .7078  | .6944  | .6811  |  |  |  |  |  |
| -1.3                                                                                  | .9680  | .9510  | .9342  | .9176  | .9012  | .8851  | .8691  | .8534  | .8379  | .8226  |  |  |  |  |  |
| -1.2                                                                                  | .11507 | .11314 | .11123 | .10933 | .10749 | .10565 | .10383 | .10204 | .10027 | .9853  |  |  |  |  |  |
| -1.1                                                                                  | .13567 | .13350 | .13136 | .12924 | .12714 | .12507 | .12302 | .12100 | .11900 | .11702 |  |  |  |  |  |
| -1.0                                                                                  | .15866 | .15625 | .15386 | .15151 | .14917 | .14686 | .14457 | .14231 | .14007 | .13786 |  |  |  |  |  |
| -0.9                                                                                  | .18406 | .18141 | .17879 | .17619 | .17361 | .17106 | .16853 | .16602 | .16354 | .16109 |  |  |  |  |  |
| -0.8                                                                                  | .21186 | .20897 | .20611 | .20327 | .20045 | .19766 | .19489 | .19215 | .18943 | .18673 |  |  |  |  |  |
| -0.7                                                                                  | .24196 | .23885 | .23576 | .23270 | .22965 | .22663 | .22363 | .22065 | .21770 | .21476 |  |  |  |  |  |
| -0.6                                                                                  | .27425 | .27093 | .26763 | .26435 | .26109 | .25785 | .25463 | .25143 | .24825 | .24510 |  |  |  |  |  |
| -0.5                                                                                  | .30854 | .30503 | .30153 | .29806 | .29460 | .29116 | .28774 | .28434 | .28096 | .27760 |  |  |  |  |  |
| -0.4                                                                                  | .34458 | .34090 | .33724 | .33360 | .32997 | .32636 | .32276 | .31918 | .31561 | .31207 |  |  |  |  |  |
| -0.3                                                                                  | .38209 | .37828 | .37448 | .37070 | .36693 | .36317 | .35942 | .35569 | .35197 | .34827 |  |  |  |  |  |
| -0.2                                                                                  | .42074 | .41683 | .41294 | .40905 | .40517 | .40129 | .39743 | .39358 | .38974 | .38591 |  |  |  |  |  |
| -0.1                                                                                  | .46017 | .45620 | .45224 | .44828 | .44433 | .44038 | .43644 | .43251 | .42858 | .42465 |  |  |  |  |  |
| -0.0                                                                                  | .50000 | .49601 | .49202 | .48803 | .48405 | .48006 | .47608 | .47210 | .46812 | .46414 |  |  |  |  |  |

| STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score. |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |
|---------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|
| Z                                                                                     | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |  |  |  |  |  |
| 0                                                                                     | .5000 | .5039 | .5078 | .5117 | .5155 | .5194 | .5232 | .5270 | .5318 | .5386 |  |  |  |  |  |
| 0.1                                                                                   | .5398 | .5438 | .5476 | .5512 | .5547 | .5582 | .5616 | .5649 | .5714 | .5735 |  |  |  |  |  |
| 0.2                                                                                   | .5792 | .5831 | .5870 | .5906 | .5943 | .5981 | .6027 | .6064 | .6102 | .6149 |  |  |  |  |  |
| 0.3                                                                                   | .6179 | .6212 | .6252 | .6290 | .6330 | .6368 | .6408 | .6441 | .6483 | .6513 |  |  |  |  |  |
| 0.4                                                                                   | .6542 | .6571 | .6626 | .6664 | .6703 | .6743 | .6774 | .6802 | .6843 | .6873 |  |  |  |  |  |
| 0.5                                                                                   | .6914 | .6947 | .6987 | .7024 | .7064 | .7084 | .7126 | .7156 | .7194 | .7240 |  |  |  |  |  |
| 0.6                                                                                   | .7257 | .7297 | .7337 | .7365 | .7391 | .7421 | .7457 | .7487 | .7519 | .7549 |  |  |  |  |  |
| 0.7                                                                                   | .7580 | .7615 | .7644 | .7670 | .7695 | .7737 | .7767 | .7795 | .7830 | .7854 |  |  |  |  |  |
| 0.8                                                                                   | .7881 | .7913 | .7938 | .7963 | .7985 | .8024 | .8051 | .8085 | .8107 | .8137 |  |  |  |  |  |
| 0.9                                                                                   | .8154 | .8189 | .8211 | .8238 | .8269 | .8294 | .8314 | .8336 | .8364 | .8391 |  |  |  |  |  |
| 1.0                                                                                   | .8413 | .8437 | .8461 | .8484 | .8508 | .8531 | .8543 | .8569 | .8595 | .8621 |  |  |  |  |  |
| 1.1                                                                                   | .8643 | .8660 | .8684 | .8706 | .8728 | .8743 | .8768 | .8790 | .8810 | .8828 |  |  |  |  |  |
| 1.2                                                                                   | .8843 | .8866 | .8877 | .8905 | .8921 | .8945 | .8967 | .8976 | .8996 | .9017 |  |  |  |  |  |
| 1.3                                                                                   | .9030 | .9049 | .9068 | .9084 | .9088 | .9149 | .9130 | .9146 | .9161 | .9174 |  |  |  |  |  |
| 1.4                                                                                   | .9194 | .9207 | .9220 | .9234 | .9257 | .9264 | .9267 | .9275 | .9292 | .9306 |  |  |  |  |  |
| 1.5                                                                                   | .9319 | .9348 | .9374 | .9399 | .9422 | .9443 | .9462 | .9479 | .9495 | .9508 |  |  |  |  |  |
| 1.6                                                                                   | .9520 | .9530 | .9538 | .9548 | .9550 | .9554 | .9559 | .9564 | .9568 | .9572 |  |  |  |  |  |
| 1.7                                                                                   | .9574 | .9583 | .9591 | .9598 | .9601 | .9604 | .9608 | .9614 | .9624 | .9637 |  |  |  |  |  |
| 1.8                                                                                   | .9640 | .9648 | .9652 | .9656 | .9658 | .9661 | .9674 | .9684 | .9696 | .9702 |  |  |  |  |  |
| 1.9                                                                                   | .9712 | .9719 | .9727 | .9732 | .9738 | .9741 | .9750 | .9753 | .9761 | .9770 |  |  |  |  |  |
| 2.0                                                                                   | .9772 | .9778 | .9783 | .9788 | .9792 | .9796 | .9800 | .9807 | .9814 | .9819 |  |  |  |  |  |
| 2.1                                                                                   | .9821 | .9827 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9853 | .9857 |  |  |  |  |  |
| 2.2                                                                                   | .9861 | .9865 | .9869 | .9871 | .9874 | .9878 | .9880 | .9884 | .9887 | .9889 |  |  |  |  |  |
| 2.3                                                                                   | .9892 | .9896 | .9898 | .9901 | .9903 | .9906 | .9907 | .9911 | .9914 | .9918 |  |  |  |  |  |
| 2.4                                                                                   | .9919 | .9923 | .9924 | .9924 | .9925 | .9926 | .9926 | .9926 | .9926 | .9926 |  |  |  |  |  |
| 2.5                                                                                   | .9937 | .9936 | .9943 | .9943 | .9943 | .9946 | .9946 | .9946 | .9946 | .9946 |  |  |  |  |  |
| 2.6                                                                                   | .9954 | .9947 | .9950 | .9953 | .9953 | .9953 | .9958 | .9960 | .9961 | .9962 |  |  |  |  |  |
| 2.7                                                                                   | .9963 | .9964 | .9964 | .9964 | .9963 | .9963 | .9970 | .9971 | .9970 | .9971 |  |  |  |  |  |
| 2.8                                                                                   | .9974 | .9972 | .9970 | .9967 | .9974 | .9974 | .9978 | .9978 | .9978 | .9980 |  |  |  |  |  |
| 2.9                                                                                   | .9981 | .9981 | .9981 | .9981 | .9981 | .9981 | .9981 | .9981 | .9981 | .9981 |  |  |  |  |  |
| 3.0                                                                                   | .9985 | .9989 | .9984 | .9988 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 |  |  |  |  |  |
| 3.1                                                                                   | .9990 | .9990 | .9991 | .9991 | .9991 | .9991 | .9991 | .9991 | .9991 | .9991 |  |  |  |  |  |
| 3.2                                                                                   | .9991 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 |  |  |  |  |  |
| 3.3                                                                                   | .9992 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 | .9993 |  |  |  |  |  |
| 3.4                                                                                   | .9996 | .9998 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 |  |  |  |  |  |
| 3.5                                                                                   | .9997 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 |  |  |  |  |  |
| 3.6                                                                                   | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 |  |  |  |  |  |
| 3.7                                                                                   | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 | .9999 |  |  |  |  |  |



**TABLE A-3** t Distribution

| $\alpha$           |                                          |                                         |                                          |                                         |                                         |                                         |
|--------------------|------------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| Degrees of Freedom | .005<br>(one tail)<br>.01<br>(two tails) | .01<br>(one tail)<br>.02<br>(two tails) | .025<br>(one tail)<br>.05<br>(two tails) | .05<br>(one tail)<br>.10<br>(two tails) | .10<br>(one tail)<br>.20<br>(two tails) | .25<br>(one tail)<br>.50<br>(two tails) |
| 1                  | 63.657                                   | 31.821                                  | 12.706                                   | 6.314                                   | 3.078                                   | 1.000                                   |
| 2                  | 9.925                                    | 6.965                                   | 4.303                                    | 2.920                                   | 1.886                                   | .816                                    |
| 3                  | 5.841                                    | 4.541                                   | 3.182                                    | 2.353                                   | 1.638                                   | .765                                    |
| 4                  | 4.604                                    | 3.747                                   | 2.776                                    | 2.132                                   | 1.533                                   | .741                                    |
| 5                  | 4.032                                    | 3.365                                   | 2.571                                    | 2.015                                   | 1.476                                   | .727                                    |
| 6                  | 3.707                                    | 3.143                                   | 2.447                                    | 1.943                                   | 1.440                                   | .718                                    |
| 7                  | 3.500                                    | 2.998                                   | 2.365                                    | 1.895                                   | 1.415                                   | .711                                    |
| 8                  | 3.355                                    | 2.896                                   | 2.306                                    | 1.860                                   | 1.397                                   | .706                                    |
| 9                  | 3.250                                    | 2.821                                   | 2.262                                    | 1.833                                   | 1.383                                   | .703                                    |
| 10                 | 3.169                                    | 2.764                                   | 2.228                                    | 1.812                                   | 1.372                                   | .700                                    |
| 11                 | 3.106                                    | 2.718                                   | 2.201                                    | 1.796                                   | 1.363                                   | .697                                    |
| 12                 | 3.054                                    | 2.681                                   | 2.179                                    | 1.782                                   | 1.356                                   | .696                                    |
| 13                 | 3.012                                    | 2.650                                   | 2.160                                    | 1.771                                   | 1.350                                   | .694                                    |
| 14                 | 2.977                                    | 2.625                                   | 2.145                                    | 1.761                                   | 1.345                                   | .692                                    |
| 15                 | 2.947                                    | 2.602                                   | 2.132                                    | 1.753                                   | 1.341                                   | .691                                    |
| 16                 | 2.921                                    | 2.584                                   | 2.120                                    | 1.746                                   | 1.337                                   | .690                                    |
| 17                 | 2.898                                    | 2.567                                   | 2.110                                    | 1.740                                   | 1.333                                   | .689                                    |
| 18                 | 2.878                                    | 2.552                                   | 2.101                                    | 1.734                                   | 1.330                                   | .688                                    |
| 19                 | 2.861                                    | 2.540                                   | 2.093                                    | 1.729                                   | 1.328                                   | .688                                    |
| 20                 | 2.845                                    | 2.528                                   | 2.086                                    | 1.725                                   | 1.325                                   | .687                                    |
| 21                 | 2.831                                    | 2.518                                   | 2.080                                    | 1.721                                   | 1.323                                   | .686                                    |
| 22                 | 2.819                                    | 2.508                                   | 2.074                                    | 1.717                                   | 1.321                                   | .686                                    |
| 23                 | 2.807                                    | 2.500                                   | 2.069                                    | 1.714                                   | 1.320                                   | .685                                    |
| 24                 | 2.797                                    | 2.492                                   | 2.064                                    | 1.711                                   | 1.318                                   | .685                                    |
| 25                 | 2.787                                    | 2.485                                   | 2.060                                    | 1.708                                   | 1.316                                   | .684                                    |
| 26                 | 2.779                                    | 2.479                                   | 2.056                                    | 1.706                                   | 1.315                                   | .684                                    |
| 27                 | 2.771                                    | 2.473                                   | 2.052                                    | 1.703                                   | 1.314                                   | .684                                    |
| 28                 | 2.763                                    | 2.467                                   | 2.048                                    | 1.701                                   | 1.313                                   | .683                                    |
| 29                 | 2.756                                    | 2.462                                   | 2.045                                    | 1.699                                   | 1.311                                   | .683                                    |
| Large (z)          | 2.575                                    | 2.326                                   | 1.960                                    | 1.645                                   | 1.282                                   | .675                                    |

| TABLE A-4          |        | Chi-Square ( $\chi^2$ ) Distribution    |        |        |        |         |         |         |         |         |
|--------------------|--------|-----------------------------------------|--------|--------|--------|---------|---------|---------|---------|---------|
|                    |        | Area to the Right of the Critical Value |        |        |        |         |         |         |         |         |
| Degrees of Freedom | 0.995  | 0.99                                    | 0.975  | 0.95   | 0.90   | 0.10    | 0.05    | 0.025   | 0.01    | 0.005   |
| 1                  | —      | —                                       | 0.001  | 0.004  | 0.016  | 2.706   | 3.841   | 5.024   | 6.635   | 7.879   |
| 2                  | 0.010  | 0.020                                   | 0.051  | 0.103  | 0.211  | 4.605   | 5.991   | 7.378   | 9.210   | 10.597  |
| 3                  | 0.072  | 0.115                                   | 0.216  | 0.352  | 0.584  | 6.251   | 7.815   | 9.348   | 11.345  | 12.838  |
| 4                  | 0.207  | 0.297                                   | 0.484  | 0.711  | 1.064  | 7.779   | 9.488   | 11.143  | 13.277  | 14.860  |
| 5                  | 0.412  | 0.554                                   | 0.831  | 1.145  | 1.610  | 9.236   | 11.071  | 12.833  | 15.086  | 16.750  |
| 6                  | 0.676  | 0.872                                   | 1.237  | 1.635  | 2.204  | 10.645  | 12.592  | 14.449  | 16.812  | 18.548  |
| 7                  | 0.989  | 1.239                                   | 1.690  | 2.167  | 2.833  | 12.017  | 14.067  | 16.013  | 18.475  | 20.278  |
| 8                  | 1.344  | 1.646                                   | 2.180  | 2.733  | 3.490  | 13.362  | 15.507  | 17.535  | 20.090  | 21.955  |
| 9                  | 1.735  | 2.088                                   | 2.700  | 3.325  | 4.168  | 14.684  | 16.919  | 19.023  | 21.666  | 23.589  |
| 10                 | 2.156  | 2.558                                   | 3.247  | 3.940  | 4.865  | 15.987  | 18.307  | 20.483  | 23.209  | 25.188  |
| 11                 | 2.603  | 3.053                                   | 3.816  | 4.575  | 5.578  | 17.275  | 19.675  | 21.920  | 24.725  | 26.757  |
| 12                 | 3.074  | 3.571                                   | 4.404  | 5.226  | 6.304  | 18.549  | 21.026  | 23.337  | 26.217  | 28.299  |
| 13                 | 3.565  | 4.107                                   | 5.009  | 5.892  | 7.042  | 19.812  | 22.362  | 24.736  | 27.688  | 29.819  |
| 14                 | 4.075  | 4.660                                   | 5.629  | 6.571  | 7.790  | 21.064  | 23.685  | 26.119  | 29.141  | 31.319  |
| 15                 | 4.601  | 5.229                                   | 6.262  | 7.261  | 8.547  | 22.307  | 24.996  | 27.488  | 30.578  | 32.801  |
| 16                 | 5.142  | 5.812                                   | 6.908  | 7.962  | 9.312  | 23.542  | 26.296  | 28.845  | 32.000  | 34.267  |
| 17                 | 5.697  | 6.408                                   | 7.564  | 8.672  | 10.085 | 24.769  | 27.587  | 30.191  | 33.409  | 35.718  |
| 18                 | 6.265  | 7.015                                   | 8.231  | 9.390  | 10.865 | 25.989  | 28.869  | 31.526  | 34.805  | 37.156  |
| 19                 | 6.844  | 7.633                                   | 8.907  | 10.117 | 11.651 | 27.204  | 30.144  | 32.852  | 36.191  | 38.582  |
| 20                 | 7.434  | 8.260                                   | 9.591  | 10.851 | 12.443 | 28.412  | 31.410  | 34.170  | 37.566  | 39.997  |
| 21                 | 8.034  | 8.897                                   | 10.283 | 11.591 | 13.240 | 29.615  | 32.671  | 35.479  | 38.932  | 41.401  |
| 22                 | 8.643  | 9.542                                   | 10.982 | 12.338 | 14.042 | 30.813  | 33.924  | 36.781  | 40.289  | 42.796  |
| 23                 | 9.260  | 10.196                                  | 11.689 | 13.091 | 14.848 | 32.007  | 35.172  | 38.076  | 41.638  | 44.181  |
| 24                 | 9.886  | 10.856                                  | 12.401 | 13.848 | 15.659 | 33.196  | 36.415  | 39.364  | 42.980  | 45.559  |
| 25                 | 10.520 | 11.524                                  | 13.120 | 14.611 | 16.473 | 34.382  | 37.652  | 40.646  | 44.314  | 46.928  |
| 26                 | 11.160 | 12.198                                  | 13.844 | 15.379 | 17.292 | 35.563  | 38.885  | 41.923  | 45.642  | 48.290  |
| 27                 | 11.808 | 12.879                                  | 14.573 | 16.151 | 18.114 | 36.741  | 40.113  | 43.194  | 46.963  | 49.645  |
| 28                 | 12.461 | 13.565                                  | 15.308 | 16.928 | 18.939 | 37.916  | 41.337  | 44.461  | 48.278  | 50.993  |
| 29                 | 13.121 | 14.257                                  | 16.047 | 17.708 | 19.768 | 39.087  | 42.557  | 45.722  | 49.588  | 52.336  |
| 30                 | 13.787 | 14.954                                  | 16.791 | 18.493 | 20.599 | 40.256  | 43.773  | 46.979  | 50.892  | 53.672  |
| 40                 | 20.707 | 22.164                                  | 24.433 | 26.509 | 29.051 | 51.805  | 55.758  | 59.342  | 63.691  | 66.766  |
| 50                 | 27.991 | 29.707                                  | 32.357 | 34.764 | 37.689 | 63.167  | 67.505  | 71.420  | 76.154  | 79.490  |
| 60                 | 35.534 | 37.485                                  | 40.482 | 43.188 | 46.459 | 74.397  | 79.082  | 83.298  | 88.379  | 91.952  |
| 70                 | 43.275 | 45.442                                  | 48.758 | 51.739 | 55.329 | 85.527  | 90.531  | 95.023  | 100.425 | 104.215 |
| 80                 | 51.172 | 53.540                                  | 57.153 | 60.391 | 64.278 | 96.578  | 101.879 | 106.629 | 112.329 | 116.321 |
| 90                 | 59.196 | 61.754                                  | 65.647 | 69.126 | 73.291 | 107.565 | 113.145 | 118.136 | 124.116 | 128.299 |
| 100                | 67.328 | 70.065                                  | 74.222 | 77.929 | 82.358 | 118.498 | 124.342 | 129.561 | 135.807 | 140.169 |

| TABLE E<br>F critical values (continued) |      |                                     |      |      |      |      |      |      |      |      |  |
|------------------------------------------|------|-------------------------------------|------|------|------|------|------|------|------|------|--|
|                                          |      | Degrees of freedom in the numerator |      |      |      |      |      |      |      |      |  |
|                                          |      | 1                                   | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    |  |
| 8                                        | 100  | 3.46                                | 3.11 | 2.93 | 2.81 | 2.71 | 2.63 | 2.57 | 2.54 | 2.56 |  |
|                                          | .050 | 5.32                                | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 |  |
|                                          | .025 | 7.57                                | 6.06 | 5.42 | 5.05 | 4.82 | 4.65 | 4.53 | 4.43 | 4.36 |  |
|                                          | .010 | 10.56                               | 8.45 | 7.57 | 6.97 | 6.60 | 6.37 | 6.24 | 6.14 | 6.07 |  |
| 9                                        | 100  | 3.41                                | 3.06 | 2.88 | 2.76 | 2.66 | 2.58 | 2.52 | 2.49 | 2.51 |  |
|                                          | .050 | 5.27                                | 4.41 | 4.02 | 3.79 | 3.64 | 3.53 | 3.45 | 3.39 | 3.34 |  |
|                                          | .025 | 7.51                                | 6.00 | 5.36 | 4.99 | 4.76 | 4.59 | 4.47 | 4.37 | 4.30 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 10                                       | 100  | 3.36                                | 3.01 | 2.83 | 2.71 | 2.61 | 2.53 | 2.47 | 2.44 | 2.46 |  |
|                                          | .050 | 5.22                                | 4.36 | 3.97 | 3.74 | 3.59 | 3.48 | 3.40 | 3.34 | 3.29 |  |
|                                          | .025 | 7.47                                | 5.96 | 5.32 | 4.95 | 4.72 | 4.55 | 4.43 | 4.33 | 4.26 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 11                                       | 100  | 3.32                                | 2.97 | 2.79 | 2.67 | 2.57 | 2.49 | 2.43 | 2.40 | 2.42 |  |
|                                          | .050 | 5.18                                | 4.32 | 3.93 | 3.70 | 3.55 | 3.44 | 3.36 | 3.30 | 3.25 |  |
|                                          | .025 | 7.43                                | 5.92 | 5.28 | 4.91 | 4.68 | 4.51 | 4.39 | 4.29 | 4.22 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 12                                       | 100  | 3.28                                | 2.93 | 2.75 | 2.63 | 2.53 | 2.45 | 2.39 | 2.36 | 2.38 |  |
|                                          | .050 | 5.14                                | 4.28 | 3.89 | 3.66 | 3.51 | 3.40 | 3.32 | 3.26 | 3.21 |  |
|                                          | .025 | 7.39                                | 5.88 | 5.24 | 4.87 | 4.64 | 4.47 | 4.35 | 4.25 | 4.18 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 13                                       | 100  | 3.24                                | 2.89 | 2.71 | 2.59 | 2.49 | 2.41 | 2.35 | 2.32 | 2.34 |  |
|                                          | .050 | 5.10                                | 4.24 | 3.85 | 3.62 | 3.47 | 3.36 | 3.28 | 3.22 | 3.17 |  |
|                                          | .025 | 7.35                                | 5.84 | 5.20 | 4.83 | 4.60 | 4.43 | 4.31 | 4.21 | 4.14 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 14                                       | 100  | 3.20                                | 2.85 | 2.67 | 2.55 | 2.45 | 2.37 | 2.31 | 2.28 | 2.30 |  |
|                                          | .050 | 5.06                                | 4.20 | 3.81 | 3.58 | 3.43 | 3.32 | 3.24 | 3.18 | 3.13 |  |
|                                          | .025 | 7.31                                | 5.80 | 5.16 | 4.79 | 4.56 | 4.39 | 4.27 | 4.17 | 4.10 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 15                                       | 100  | 3.16                                | 2.81 | 2.63 | 2.51 | 2.41 | 2.33 | 2.27 | 2.24 | 2.26 |  |
|                                          | .050 | 5.02                                | 4.16 | 3.77 | 3.54 | 3.39 | 3.28 | 3.20 | 3.14 | 3.09 |  |
|                                          | .025 | 7.27                                | 5.76 | 5.12 | 4.75 | 4.52 | 4.35 | 4.23 | 4.13 | 4.06 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 16                                       | 100  | 3.12                                | 2.77 | 2.59 | 2.47 | 2.37 | 2.29 | 2.23 | 2.20 | 2.22 |  |
|                                          | .050 | 5.00                                | 4.14 | 3.75 | 3.52 | 3.37 | 3.26 | 3.18 | 3.12 | 3.07 |  |
|                                          | .025 | 7.23                                | 5.72 | 5.08 | 4.71 | 4.48 | 4.31 | 4.19 | 4.09 | 4.02 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |
| 17                                       | 100  | 3.08                                | 2.73 | 2.55 | 2.43 | 2.33 | 2.25 | 2.19 | 2.16 | 2.18 |  |
|                                          | .050 | 4.96                                | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.08 | 3.03 |  |
|                                          | .025 | 7.19                                | 5.68 | 5.04 | 4.67 | 4.44 | 4.27 | 4.15 | 4.05 | 3.98 |  |
|                                          | .010 | 10.54                               | 8.43 | 7.55 | 6.95 | 6.58 | 6.35 | 6.22 | 6.12 | 6.05 |  |

| TABLE E<br>F critical values (continued) |      |                                     |      |      |      |      |      |      |      |      |      |      |
|------------------------------------------|------|-------------------------------------|------|------|------|------|------|------|------|------|------|------|
|                                          |      | Degrees of freedom in the numerator |      |      |      |      |      |      |      |      |      |      |
|                                          |      | 10                                  | 12   | 15   | 20   | 25   | 30   | 40   | 50   | 60   | 120  | 1000 |
| 11                                       | 100  | 2.54                                | 2.50 | 2.46 | 2.43 | 2.40 | 2.38 | 2.36 | 2.35 | 2.34 | 2.33 | 2.30 |
|                                          | .050 | 3.35                                | 3.28 | 3.22 | 3.15 | 3.11 | 3.08 | 3.04 | 3.02 | 3.01 | 2.97 | 2.93 |
|                                          | .025 | 4.30                                | 4.20 | 4.10 | 4.00 | 3.94 | 3.89 | 3.84 | 3.81 | 3.78 | 3.73 | 3.68 |
|                                          | .010 | 5.77                                | 5.67 | 5.56 | 5.45 | 5.37 | 5.30 | 5.24 | 5.19 | 5.14 | 5.07 | 4.96 |
| 12                                       | 100  | 2.52                                | 2.48 | 2.44 | 2.41 | 2.38 | 2.36 | 2.34 | 2.33 | 2.32 | 2.31 | 2.29 |
|                                          | .050 | 3.33                                | 3.26 | 3.20 | 3.13 | 3.09 | 3.06 | 3.02 | 3.00 | 2.99 | 2.95 | 2.91 |
|                                          | .025 | 4.28                                | 4.18 | 4.08 | 3.98 | 3.92 | 3.87 | 3.82 | 3.79 | 3.76 | 3.72 | 3.66 |
|                                          | .010 | 5.75                                | 5.65 | 5.54 | 5.43 | 5.35 | 5.28 | 5.22 | 5.17 | 5.12 | 5.05 | 4.94 |
| 13                                       | 100  | 2.50                                | 2.46 | 2.42 | 2.39 | 2.36 | 2.34 | 2.32 | 2.31 | 2.30 | 2.29 | 2.27 |
|                                          | .050 | 3.31                                | 3.24 | 3.18 | 3.11 | 3.07 | 3.04 | 3.00 | 2.98 | 2.97 | 2.93 | 2.89 |
|                                          | .025 | 4.26                                | 4.16 | 4.06 | 3.96 | 3.90 | 3.85 | 3.80 | 3.77 | 3.74 | 3.69 | 3.64 |
|                                          | .010 | 5.73                                | 5.63 | 5.52 | 5.41 | 5.33 | 5.26 | 5.20 | 5.15 | 5.10 | 5.03 | 4.92 |
| 14                                       | 100  | 2.48                                | 2.44 | 2.40 | 2.37 | 2.34 | 2.32 | 2.30 | 2.29 | 2.28 | 2.27 | 2.25 |
|                                          | .050 | 3.29                                | 3.22 | 3.16 | 3.09 | 3.05 | 3.02 | 2.98 | 2.96 | 2.95 | 2.91 | 2.87 |
|                                          | .025 | 4.24                                | 4.14 | 4.04 | 3.94 | 3.88 | 3.83 | 3.78 | 3.75 | 3.72 | 3.67 | 3.62 |
|                                          | .010 | 5.71                                | 5.61 | 5.50 | 5.39 | 5.31 | 5.24 | 5.18 | 5.13 | 5.08 | 5.01 | 4.90 |
| 15                                       | 100  | 2.46                                | 2.42 | 2.38 | 2.35 | 2.32 | 2.30 | 2.28 | 2.27 | 2.26 | 2.25 | 2.23 |
|                                          | .050 | 3.27                                | 3.20 | 3.14 | 3.07 | 3.03 | 3.00 | 2.96 | 2.94 | 2.93 | 2.89 | 2.85 |
|                                          | .025 | 4.22                                | 4.12 | 4.02 | 3.92 | 3.86 | 3.81 | 3.76 | 3.73 | 3.70 | 3.65 | 3.60 |
|                                          | .010 | 5.69                                | 5.59 | 5.48 | 5.37 | 5.29 | 5.22 | 5.16 | 5.11 | 5.06 | 4.99 | 4.88 |
| 16                                       | 100  | 2.44                                | 2.40 | 2.36 | 2.33 | 2.30 | 2.28 | 2.26 | 2.25 | 2.24 | 2.23 | 2.21 |
|                                          | .050 | 3.25                                | 3.18 | 3.12 | 3.05 | 3.01 | 2.98 | 2.94 | 2.92 | 2.91 | 2.87 | 2.83 |
|                                          | .025 | 4.20                                | 4.10 | 4.00 | 3.90 | 3.84 | 3.79 | 3.74 | 3.71 | 3.68 | 3.63 | 3.58 |
|                                          | .010 | 5.67                                | 5.57 | 5.46 | 5.35 | 5.27 | 5.20 | 5.14 | 5.09 | 5.04 | 4.97 | 4.86 |
| 17                                       | 100  | 2.42                                | 2.38 | 2.34 | 2.31 | 2.28 | 2.26 | 2.24 | 2.23 | 2.22 | 2.21 | 2.19 |
|                                          | .050 | 3.23                                | 3.16 | 3.10 | 3.03 | 2.99 | 2.96 | 2.92 | 2.90 | 2.89 | 2.85 | 2.81 |
|                                          | .025 | 4.18                                | 4.08 | 3.98 | 3.88 | 3.82 | 3.77 | 3.72 | 3.69 | 3.66 | 3.61 | 3.56 |
|                                          | .010 | 5.65                                | 5.55 | 5.44 | 5.33 | 5.25 | 5.18 | 5.12 | 5.07 | 5.02 | 4.95 | 4.84 |
| 18                                       | 100  | 2.40                                | 2.36 | 2.32 | 2.29 | 2.26 | 2.24 | 2.22 | 2.21 | 2.20 | 2.19 | 2.17 |
|                                          | .050 | 3.21                                | 3.14 | 3.08 | 3.01 | 2.97 | 2.94 | 2.90 | 2.88 | 2.87 | 2.83 | 2.79 |
|                                          | .025 | 4.16                                | 4.06 | 3.96 | 3.86 | 3.80 | 3.75 | 3.70 | 3.67 | 3.64 | 3.59 | 3.54 |
|                                          | .010 | 5.63                                | 5.53 | 5.42 | 5.31 | 5.23 | 5.16 | 5.10 | 5.05 | 5.00 | 4.93 | 4.82 |
| 19                                       | 100  | 2.38                                | 2.34 | 2.30 | 2.27 | 2.24 | 2.22 | 2.20 | 2.19 | 2.18 | 2.17 | 2.15 |
|                                          | .050 | 3.19                                | 3.12 | 3.06 | 2.99 | 2.95 | 2.92 | 2.88 | 2.86 | 2.85 | 2.81 | 2.77 |
|                                          | .025 | 4.14                                | 4.04 | 3.94 | 3.84 | 3.78 | 3.73 | 3.68 | 3.65 | 3.62 | 3.57 | 3.52 |
|                                          | .010 | 5.61                                | 5.51 | 5.40 | 5.29 | 5.21 | 5.14 | 5.08 | 5.03 | 4.98 | 4.91 | 4.80 |
| 20                                       | 100  | 2.36                                | 2.32 | 2.28 | 2.25 | 2.22 | 2.20 | 2.18 | 2.17 | 2.16 | 2.15 | 2.13 |
|                                          | .050 | 3.17                                | 3.10 | 3.04 | 2.97 | 2.93 | 2.90 | 2.86 | 2.84 | 2.83 | 2.79 | 2.75 |
|                                          | .025 | 4.12                                | 4.02 | 3.92 | 3.82 | 3.76 | 3.71 | 3.66 | 3.63 | 3.60 | 3.55 | 3.50 |
|                                          | .010 | 5.59                                | 5.49 | 5.38 | 5.27 | 5.19 | 5.12 | 5.06 | 5.01 | 4.96 | 4.89 | 4.78 |
| 22                                       | 100  | 2.32                                | 2.28 | 2.24 | 2.21 | 2.18 | 2.16 | 2.14 | 2.13 | 2.12 | 2.11 | 2.08 |
|                                          | .050 | 3.13                                | 3.06 | 3.00 | 2.93 | 2.89 | 2.86 | 2.82 | 2.80 | 2.79 | 2.75 | 2.71 |
|                                          | .025 | 4.08                                | 3.98 | 3.88 | 3.78 | 3.72 | 3.67 | 3.62 | 3.59 | 3.56 | 3.51 | 3.46 |
|                                          | .010 | 5.57                                | 5.47 | 5.36 | 5.25 | 5.17 | 5.10 | 5.04 | 4.99 | 4.94 | 4.87 | 4.76 |
| 24                                       | 100  | 2.28                                | 2.24 | 2.20 | 2.17 | 2.14 | 2.12 | 2.10 | 2.09 | 2.08 | 2.06 | 2.04 |
|                                          | .050 | 3.09                                | 3.02 | 2.96 | 2.89 | 2.85 | 2.82 | 2.78 | 2.76 | 2.75 | 2.71 | 2.67 |
|                                          | .025 | 4.04                                | 3.94 | 3.84 | 3.74 | 3.68 | 3.63 | 3.58 | 3.55 | 3.52 | 3.47 | 3.42 |
|                                          | .010 | 5.55                                | 5.45 | 5.34 | 5.23 | 5.15 | 5.08 | 5.02 | 4.97 | 4.92 | 4.85 | 4.74 |
| 26                                       | 100  | 2.24                                | 2.20 | 2.16 | 2.13 | 2.10 | 2.08 | 2.06 | 2.05 | 2.04 | 2.02 | 2.00 |
|                                          | .050 | 3.05                                | 2.98 | 2.92 | 2.85 | 2.81 | 2.78 | 2.74 | 2.72 | 2.71 | 2.67 | 2.63 |
|                                          | .025 | 3.98                                | 3.88 | 3.78 | 3.68 | 3.62 | 3.57 | 3.52 | 3.49 | 3.46 | 3.41 | 3.36 |
|                                          | .010 | 5.53                                | 5.43 | 5.32 | 5.21 | 5.13 | 5.06 | 5.00 | 4.95 | 4.90 | 4.83 | 4.72 |
| 28                                       | 100  | 2.20                                | 2.16 | 2.12 | 2.09 | 2.06 | 2.04 | 2.02 | 2.01 | 2.00 | 1.98 | 1.96 |
|                                          | .050 | 3.01                                | 2.94 | 2.88 | 2.81 | 2.77 | 2.74 | 2.70 | 2.68 | 2.67 | 2.63 | 2.59 |
|                                          | .025 | 3.94                                | 3.84 | 3.74 | 3.64 | 3.58 | 3.53 | 3.48 | 3.45 | 3.42 | 3.37 | 3.32 |
|                                          | .010 | 5.49                                | 5.39 | 5.28 | 5.17 | 5.09 | 5.02 | 4.96 | 4.91 | 4.86 | 4.79 | 4.68 |
| 30                                       | 100  | 2.18                                | 2.14 | 2.10 | 2.07 | 2.04 | 2.02 | 2.00 | 1.99 | 1.98 | 1.96 | 1.94 |
|                                          | .050 | 2.99                                | 2.92 | 2.86 | 2.79 | 2.75 | 2.72 | 2.68 | 2.66 | 2.65 | 2.61 | 2.57 |
|                                          | .025 | 3.92                                | 3.82 | 3.72 | 3.62 | 3.56 | 3.51 | 3.46 | 3.43 | 3.40 | 3.35 | 3.30 |
|                                          | .010 | 5.47                                | 5.37 | 5.26 | 5.15 | 5.07 | 5.00 | 4.94 | 4.89 | 4.84 | 4.77 | 4.66 |